Movement and kinematics

Introduction

Kinematics is the branch of physics that studies **how objects move**, without worrying about why they move.

Example: when you ride a bicycle, you can pay attention to **where you are, what path you follow, how fast you are going, or whether you are accelerating**. All of that is studied by kinematics.

Motion is relative

What does this mean? A person sitting next to you on the bus is at rest from your point of view. But from the perspective of someone sitting at the bus stop, that person is moving.

> **!** Very important
>
> Motion always depends on the observer: something moves when its position changes with respect to a point that we choose (in this case, relative to the observer). This point is called the **reference point** or **origin of the reference system**.

Figure 1: Bus stop

Reference system and position

Imagine you are talking to a friend so they can find you in a park, and you tell them you are one hundred meters from the entrance. With that information, you have given a point that acts as the origin of the reference system ("the entrance") and the position ("one hundred meters from that entrance").

Figure 2: Sistema referencia 1D

> ❗ Very important
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> - **Reference system**: a set of axes and a point, which is the origin, from which we measure positions.
>   - In each problem we must choose a reference system and it will be considered fixed and still.
> - **Position (x)**: the place where an object is with respect to the origin of the reference system. Unit: **meter (m)**.

We say that a "body" is in **motion** when its position relative to the origin of the reference frame changes over time.

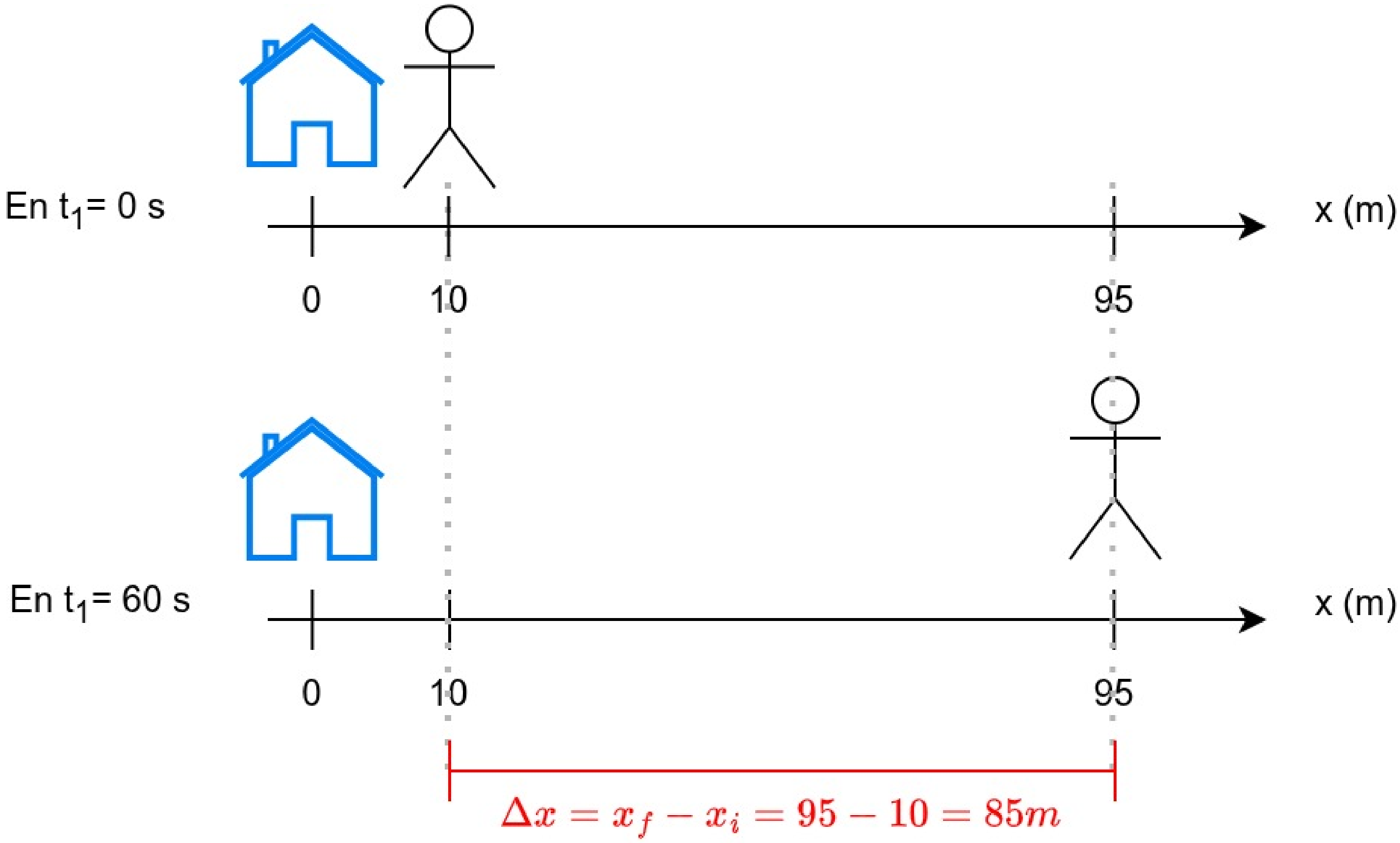

Figure 3: Motion along the horizontal axis with a reference frame

Trajectory

Trajectory: is the path an object follows, that is, the set of its positions over time. For example: the points a cyclist would pass through following the road in the drawing.

Figure 4: Trajectory

Trajectories can be straight (a ball we drop or a car on a straight road) or curved like the previous image or the shot of a ball into a basket.

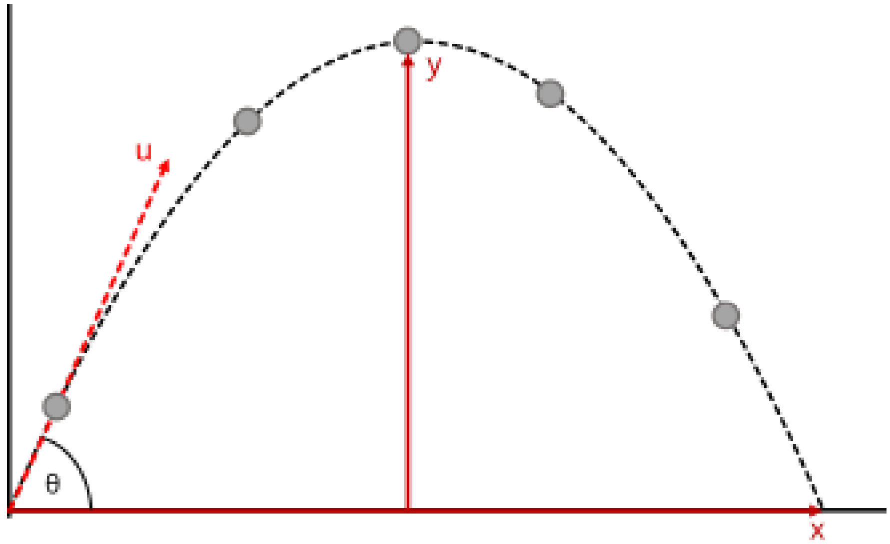

Figure 5: Curved trajectory of a ball when shooting a basket

Exercises

1. Define trajectory in your own words.
2. Can the position change without changing the trajectory?
3. Give an example of a curved trajectory.
4. How would you draw the trajectory of a person who wants to go from the school to the sports field?

Soluciones

1. The path an object follows.
2. Yes, if it moves along the same type of trajectory.
3. A thrown ball.
4. This exercise has different solutions. The interesting thing is simply to consider that it must pass through numerous streets.

Distance traveled and displacement

Very important

Distance traveled (s): This is the total distance traveled by an object during its motion. Unit: **meter (m)**.
Displacement (Δx): This is the distance between the initial position (x_i) and the final position (x_f). It is also measured in **meters (m)** and is calculated as: $\Delta x = x_f - x_i$

When we travel in a straight line and don't backtrack, the distance traveled is equal to the displacement.

But when we go somewhere and return, these values may not match. For example, sometimes the distance traveled might be 100 m, but the displacement is 0. When? When we go from our house to the school, which is 50 m away, and then return.

Exercises

1. Is distance traveled the same as displacement?
2. Calculate the distance traveled if you go from your house to school (200 m), from there to the supermarket (another 60 m in the same direction), and from the supermarket back home (260 m from the supermarket to home in the opposite direction).
3. Now calculate the displacement if you go to the places mentioned above, considering your house as the starting and ending point.
4. What would the displacement be if you went from home to school and from there to the supermarket without returning home?
5. And the distance traveled?
6. Can the distance traveled be zero?

Soluciones

1. No.
2. $s = 200 + 60 + 260 = 520m$.
3. $\Delta x = x_f - x_i = 0 - 0 = 0m$.
4. $\Delta x = x_f - x_i = (200 + 60) - 0 = 260m$.
5. $s = 200 + 60 = 260m$.
6. Yes, only when there is no movement, that is, if you are not moving.

Speed and velocity

Going fast (speed) is not the same as knowing where you're going at that speed (velocity).

- **Speed**: how fast something is moving. Unit: **m/s**.
- **Velocity**: speed + direction. It is represented by an arrow. The length of that arrow, which is the magnitude, is the speed.

! Very important

It is very important to know how to calculate speed, and also how to solve for all the variables in the formula.

- Speed: $v = \frac{s}{t}$. It is measured in **meters per second ($\frac{m}{s}$)**
- Solving for distance from speed: $v = \frac{s}{t} \rightarrow v \cdot t = s$. It is measured in **meters (m)**.

- Solving for time from speed: $v = \frac{s}{t} \rightarrow v \cdot t = s \rightarrow t = \frac{s}{v}$. It is measured in **seconds (s)**.